Fractional Knapsack

Topic: Greedy File: g2.pdf

PROBLEM

Each item has a weight and value. Maximize total value in a capacity-limited bag; fractions of items are allowed.

GREEDY CHOICE

Compute value / weight for every item, sort by this ratio descending, then take as much as possible from each item in that order.

PSEUDOCODE

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sort items by value/weight descending
for each item:
    take min(item weight, remaining capacity)
    add its proportional value
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EXAMPLE

Capacity 50; items (weight,value) are (10,60), (20,100), (30,120). Take the first two and 20 of the final item's 30 weight; value = 240.

COMPLEXITY

Sorting: O(n log n); selection scan: O(n). Overall time: O(n log n).

IMPORTANT

This greedy rule is optimal for fractional knapsack, but not generally for 0/1 knapsack where items cannot be split.